	Raffles Institution Physics Department Revision Package 3	**Motion & Forces**	**3**

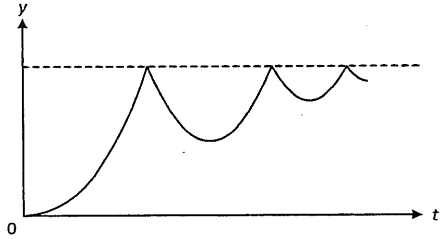
Important note: ****denotes questions that may require knowledge from later chapters in order to solve them.***

I. MCQs (ACJC – YJC)

1. **[ACJC/H2/Prelims 2018/P1/3]**

A ball is released from rest above a horizontal surface and bounces several times.

The graph shows the variation with time of quantity y for the ball.



What is quantity y?

A displacement from horizontal surface; taking upwards as positive

B displacement from horizontal surface; taking downwards as positive

C displacement from where ball is released; taking upwards as positive

D displacement from where ball is released; taking downwards as positive

2. **[ACJC/H2/Prelims 2018/P1/4]**

A student, standing on the platform of an MRT train station, notices that the first two carriages of an arriving train passes her in 2.00 s, and the next two carriages in another 2.40 s. The train is decelerating uniformly and each carriage is 20.0 m long.

What is the deceleration of the MRT train?

A 1.38 m s^{-2} B 1.52 m s^{-2} C 1.67 m s^{-2} D 3.33 m s^{-2}

3. **[AJC/H2/Prelims 2018/P1/2]**

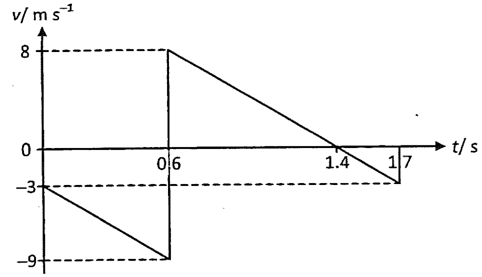
A cannon at the top of a 30 m high hill fires a shell at an angle of 30° upwards from the horizontal with a speed of 50 m s^{-1}.

Taking air resistance to be negligible, what is the angle to the vertical at which the shell lands on level ground?

A 39° B 42° C 48° D 51°

4. **[CJC/H2/Prelims 2018/P1/3]**

A student throws a rubber ball vertically downwards at a speed of 3.0 m s^{-1}. It hits the ground and rebounds vertically. The graph below shows the velocity–time graph for the first 1.7 s of the motion of the rubber ball

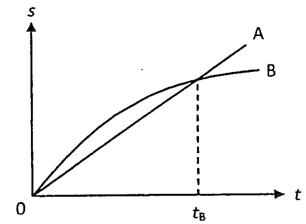


What is the displacement of the ball between the point at which it was first thrown and the highest point of the motion?

A zero B 0.4 m C 1.8 m D 3.6 m

5. **[EJC/H2/Prelims 2018/P1/2 Modified]**

The diagram below shows the variation of displacement s with time t for Train A and Train B, running on parallel tracks.



Which of the following statements is correct?

A At time t_B, both trains have the same velocity.

B Both trains speed up all the time.

C Both trains have the same velocity at some time before t_B.

D Somewhere on the graph, both trains have the same acceleration.

6. **[HCI/H2/Prelims 2018/P1/2]**

An elevator is moving downwards with an acceleration of 5.8 m s^{-2}. A ball, held 2.0 m above the floor of the elevator and at rest with respect to the elevator, is released.

How long does it take for the ball to reach the floor of the elevator?

A 0.51 s B 0.64 s C 0.83 s D 1.00 s

7. **[MI/H2/Prelims 2018/P1/3]**

An e-scooter travels at a uniform velocity of 15 m s^{-1} along a level road and passes a motorised bicycle which is initially at rest at $t = 0$ s. The motorised bicycle immediately accelerates uniformly along the same direction to catch up with the e-scooter. At $t = 6$ s, both the e-scooter and the motorised bicycle are 54 m apart.

What additional time will it take for the motorised bicycle to catch up with the e-scooter?

A 4.6 s **B** 9.0 s **C** 10.0 s **D** 15.0 s

8. **[NYJC/H2/Prelims 2018/P1/3]**

A train initially at rest accelerates at a constant rate of 1.4 m s^{-2} for 10 s and then slows down at a constant rate of 0.20 m s^{-2} until it comes to a rest. The total distance travelled by the train is

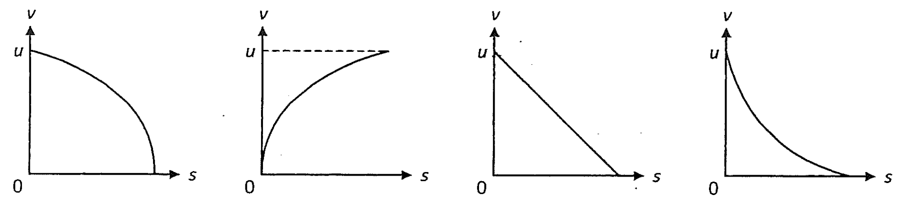
A 80 m **B** 220 m **C** 560 m **D** 1120 m

9. **[NYJC/H2/Prelims 2018/P1/4]**

A falling stone strikes some soft ground at speed u and suffers a constant deceleration until it stops.

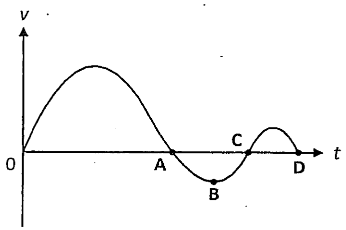
Which one of the following graphs best represents the variation with distance s of the stone's speed, measured downwards, from the surface of the ground?

A **B** **C** **D**



10. **[PJC/H2/Prelims 2018/P1/2 Modified]**

A graph showing the variation with time t of the velocity v of a body is as shown.

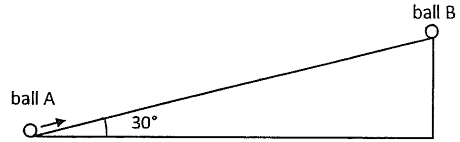


At which point on the graph is the body furthest away from its starting position?

11. **[RVHS/H2/Prelims 2018/P1/2]**

Ball A is launched up the slope from the bottom of a smooth inclined plane with an initial speed u.

At the same time that ball A is launched, ball B is released from rest at the top of the inclined plane. The two balls meet at the mid-point of the inclined plane after two seconds.



Given that the mass of ball A is twice that of ball B, what is the speed u?

A 4.9 m s^{-1}

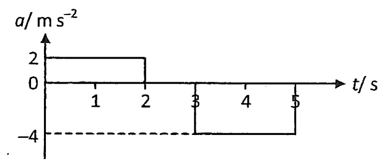
B 9.8 m s^{-1}

C 17 m s^{-1}

D 20 m s^{-1}

12. **[SRJC/H2/Prelims 2018/P1/4]**

The graph below represents the variation with time t of the acceleration a of a car starting from rest.

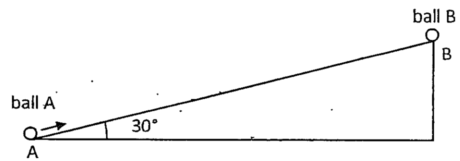


What is the total displacement of the car from the starting point at the end of 5 s?

A – 4 m **B** 4 m **C** 8 m **D** 12 m

13. **[VJC/H2/Prelims 2018/P1/2]**

Ball A is launched up an inclined plane from point A with an initial speed that is the minimum speed for it to just reach point B at the top of the plane. At the same moment that ball B is launched up the plane, ball B is released from rest from point B. The two balls collide at a point C somewhere on the inclined plane.



What is the ratio of the distance AC to the distance BC?

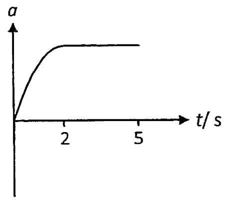
A 1 **B** 2 **C** 3 **D** 4

14. [ACJC/H2/Prelims 2018/P1/5 Modified]

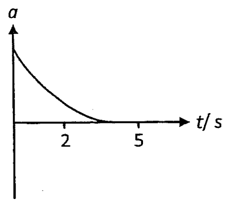
A parachutist steps out from an aircraft, falls for 2 s and then opens his parachute.

Which graph best represents the variation with time t of his vertical acceleration a during the first 5 s of his descent?

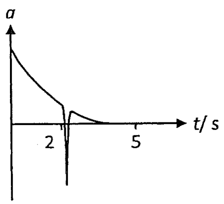
A



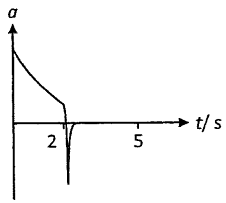
B



C



D



15. [AJC/H2/Prelims 2018/P1/3]

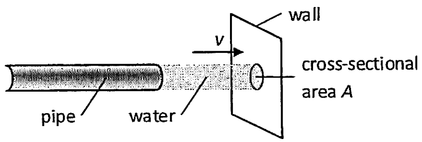
A model helicopter of mass 5.0 kg rises with constant acceleration from rest to a height of 60 m in 10 s.

What is the upward force exerted on the model by the air?

A 3.0 N B 6.0 N C 43 N D 55 N

16. [DHS/H2/Prelims 2018/P1/5]

Water flows out of a pipe and hits a wall.



When the jet of water hits the wall, it has horizontal velocity v and cross-sectional area A.

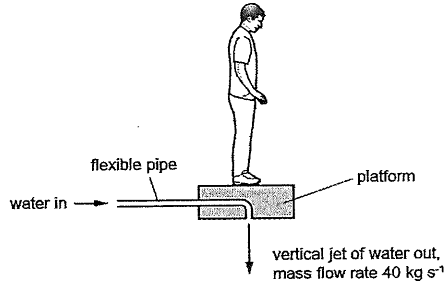
The density of the water is ρ. The water does not rebound from the wall.

What is the force exerted on the wall by the water?

A $\dfrac{\rho v}{A}$ B $\dfrac{\rho v^2}{A}$ C $A\rho v$ D $A\rho v^2$

17. [EJC/H2/Prelims 2018/P1/4]

The diagram shows a man standing on a platform that is attached to a flexible pipe. Water is pumped through the pipe so that the man and platform remain at a constant height.



The resultant vertical force on the platform is zero. The combined mass of the man and platform is 96 kg. The mass of water that is discharged vertically downwards from the platform each second is 40 kg.

What is the speed of the water leaving the platform?

A 2.4 m s^{-1} B 6.9 m s^{-1} C 24 m s^{-1} D 47 m s^{-1}

18. [HCI/H2/Prelims 2018/P1/5]

An empty truck has a mass of 5000 kg. Regardless of the mass of load it has, it experiences a fixed retarding force of 70000 N when it decelerates from a speed of 50 m s^{-1} to 30 m s^{-1}. The duration for the empty truck to decelerate is t_1. The duration for it to decelerate when it has a full load of 1300 kg is t_2.

What is the difference $(t_2 - t_1)$?

A 0.37 s B 0.56 s C 0.93 s D 1.43 s

19. [IJC/H2/Prelims 2018/P1/4]

A firework rocket is fired vertically upwards. The fuel burns and produces a constant upwards force on the rocket. After 5 seconds, there is no fuel left. Air resistance is negligible.

What is the acceleration before and after 5 seconds?

	before 5 seconds	after 5 seconds
A	constant	constant
B	constant	zero
C	increasing	constant
C	increasing	zero

20. [MI/H2/Prelims 2018/P1/4]

A squash ball approaches a squash player who gives it a hard hit with a swing of her racket. The ball then returns directly to her opponent.

Which of the following is true about the squash ball and the racket during the time of contact?

A The force on the racket due to the ball is smaller than the force on the ball due to the racket because the ball is much smaller and lighter than the racket.

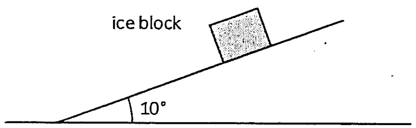
B The force on the racket due to the ball is smaller than the force on the ball due to the racket because the ball moves off with high speed after the contact whereas the racket does not.

C The force on the racket due to the ball is larger than the force on the ball due to the racket because smaller mass means smaller force.

D The force on the racket due to the ball is equal to the force on the ball due to the racket.

21. [MI/H2/Prelims 2018/P1/5]

A block of ice of mass 1.8 kg slides down a smooth slope from rest.



What is the momentum of the ice block after 3.0 s?

A $9.2\ \text{kg m s}^{-1}$ B $18.4\ \text{kg m s}^{-1}$ C $27.2\ \text{kg m s}^{-1}$ D $53.0\ \text{kg m s}^{-1}$

22. [MJC/H2/Prelims 2018/P1/4]

A body of mass 3.0 kg is thrown with a velocity of $20\ \text{m s}^{-1}$ at an angle of 60° above horizontal. It reaches the maximum height after 1.8 s. Air resistance is negligible.

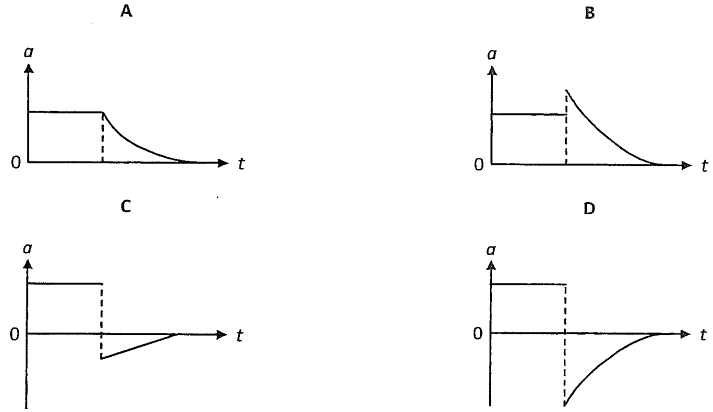
What is the rate of change of momentum of the body at the maximum height?

A zero B $17\ \text{kg m s}^{-2}$ C $29\ \text{kg m s}^{-2}$ D $33\ \text{kg m s}^{-2}$

23. [NJC/H2/Prelims 2018/P1/3]

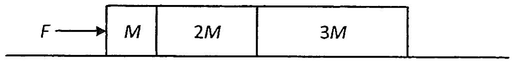
A small metal sphere is released from rest at 1 m above the surface of a viscous fluid. The sphere experiences a viscous force that is proportional to its velocity when moving in the fluid.

Which of the following graphs best represents the variation of the acceleration a of the sphere with time t?



24. [NYJC/H2/Prelims 2018/P1/6]

Three blocks with masses M, $2M$ and $3M$ are pushed along a rough surface by a horizontal force F as shown.



The frictional force between a block and the rough surface is directly proportional to the weight of the block.

Given that the frictional force between the block with mass M and the rough surface is f, what is the magnitude of the force mass $3M$ exerts on mass $2M$?

A $\frac{1}{2}F$

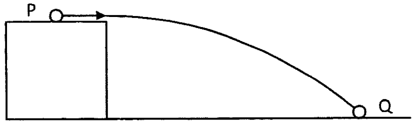
B $\frac{1}{2}F+3f$

C $\frac{1}{3}F+f$

D $\frac{1}{3}F+3f$

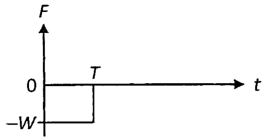
25. [RVHS/H2/Prelims 2018/P1/4]

A ball of weight W moves along a smooth horizontal surface until it falls off the edge at time T.

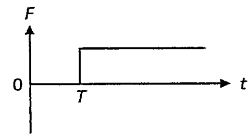


Given that air resistance is not negligible, which of the following graphs shows the variation with time t of the resultant vertical force F acting on the ball as it moves from P to Q?

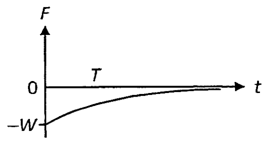
A



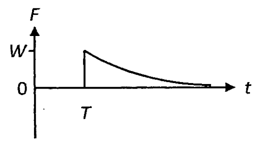
B



C

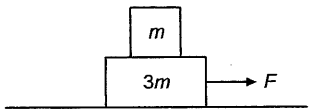


D



26. [TPJC/H2/Prelims 2018/P1/8 Modified]

A box of mass m rests on another box of mass $3m$. The more massive box slides along a frictionless surface due to a pulling force F.

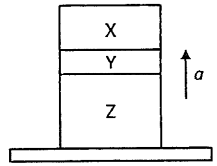


What is the friction between the boxes?

A F **B** $\frac{1}{2}F$ **C** $\frac{1}{3}F$ **D** $\frac{1}{4}F$

27. [VJC/H2/Prelims 2018/P1/4]

Three crates X, Y and Z of masses $3m$, m and $5m$ respectively, are stacked on top of one another on the floor of a lift which is moving upwards with an acceleration a.



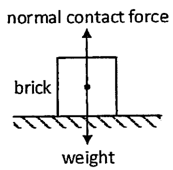
What is the magnitude of the force exerted by crate Y on crate Z during this acceleration?

A $4mg+4ma$ **B** $4mg+5ma$ **C** $4mg-4ma$ **D** $4mg-5ma$

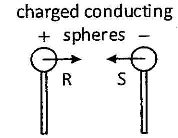
28. [JJC/H2/Prelims 2018/P1/4]

Which diagram shows a pair of forces that are not related to one another by Newton's third law?

A

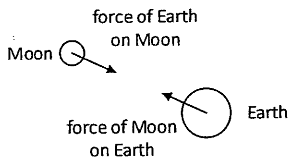


B

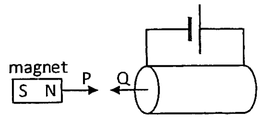


R = force on positive charge
S = force on negative charge

C



D



P = force of solenoid on magnet
Q = force of magnet on solenoid

II. Structured Questions (ACJC – YJC)

1. **[AJC/H2/Prelims 2018/P2/1 Modified]**

Ball A falls vertically in air from rest. The variation with time t of the distance d moved by the ball is shown in Fig. 1.1.

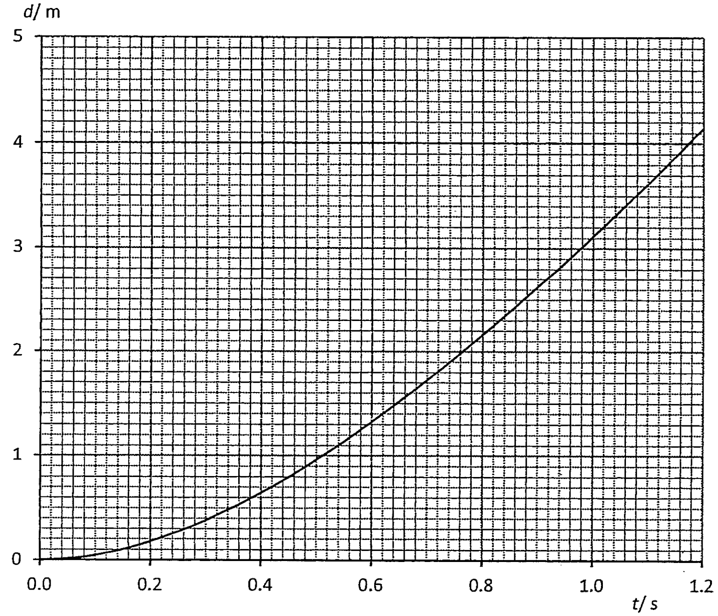


Fig. 1.1

*a. By reference to Fig. 1.1, explain how it can be deduced that air resistance is not negligible. [2]

b. Use Fig. 1.1 to determine

i. the speed of the ball at a time of 0.40 s after it has been released, [2]

ii. the average speed of the ball in the first 0.40 s after it has been released. [1]

c. Ball A is replaced by ball B which experiences negligible air resistance.

i. Calculate the distance travelled by ball B after falling for 0.40 s from rest. [1]

ii. On Fig. 1.1, sketch a graph to show the variation with time t of the distance d moved by ball B after falling from rest. Label the graph P. [3]

[Total: 9]

2. **[DHS/H2/Prelims 2018/P3/2]**

A hot-air balloon rises vertically at a constant speed. At time $t = 0$, a ball is released from the balloon. Fig. 2.1 shows the variation with time t of the ball's velocity v.

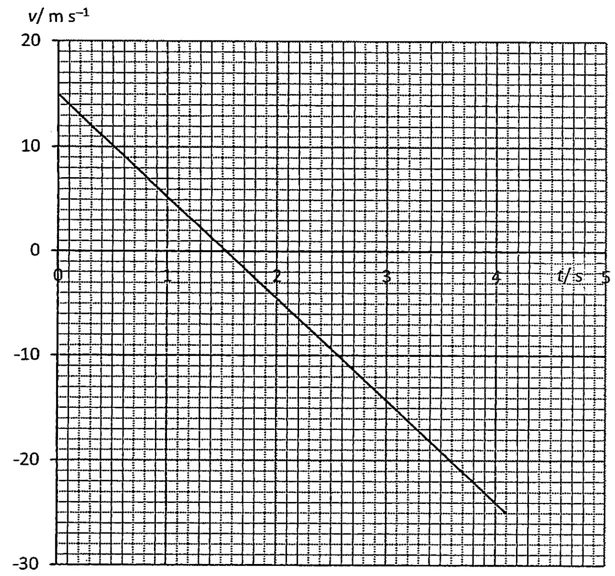


Fig. 2.1

The ball hits the ground at t = 4.1 s.

a. State the speed of the hot-air balloon. [1]

b. Explain how the graph in Fig. 2.1 shows that the acceleration of the ball is constant. [1]

c. Use Fig. 2.1 to

i. state the time at which the ball reaches its highest point, [1]

ii. show that the ball rises for a further 12 m between release and its highest point, [1]

iii. determine the distance between the point of release of the ball and the ground. [2]

d. Describe the difference between displacement of the ball and the distance it travels. [2]

*e. Sketch a new graph on Fig. 2.1 showing the variation with t of the ball's velocity v if air resistance is not negligible. Assume terminal velocity is attained by the ball before hitting the ground. Label the new graph N. [2]

[Total: 10]

3. **[JJC/H2/Prelims 2018/P2/1]**

A tennis ball is thrown vertically downwards from a height of 0.65 m above the ground. It leaves the hand with an initial speed of 1.5 m s^{-1}. The ball rebounds and is caught when it is travelling upwards with a speed of 1.0 m s^{-1}. Assume that air resistance is negligible.

a. Calculate the speed of the ball just before it strikes the ground. [2]

b. The tennis ball is thrown downwards at $t = 0$. It hits the ground at time t_1 and is caught at time t_2.

On Fig. 3.1, sketch the velocity-time graph for the motion of the ball from the time it leaves the hand to when it returns. Assume that the contact time between the ball and the ground is negligible. The initial velocity **X** and final velocity **Y** are marked on Fig. 3.1. [2]

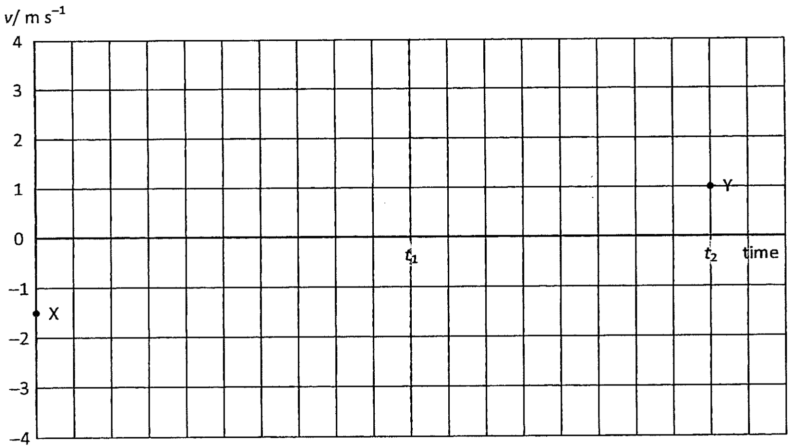


Fig. 3.1

*c. State and explain whether the bounce is elastic. [2]

*d. On Fig. 3.1, sketch the velocity-time graph of the tennis ball if air resistance is not negligible. Label this graph **P**. [2]

[Total: 8]

4. **[TPJC/H2/Prelims 2018/P3/1 Modified]**

An object of mass 1.5 kg is released from a stationary hot air balloon. Fig. 4.1 shows how the velocity of the object varies with time.

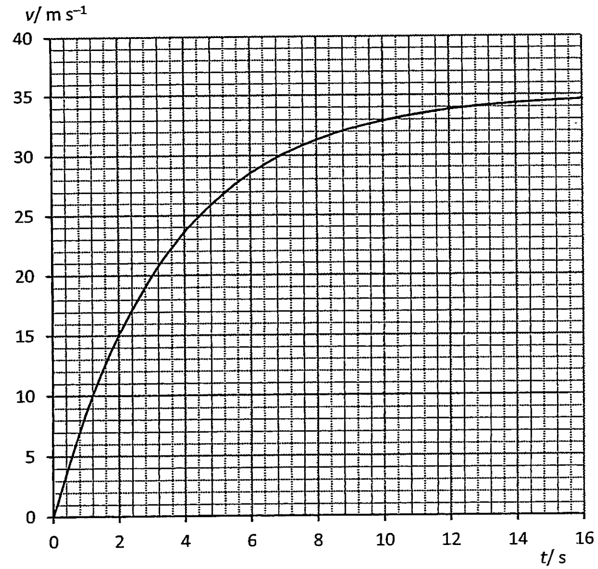


Fig. 4.1

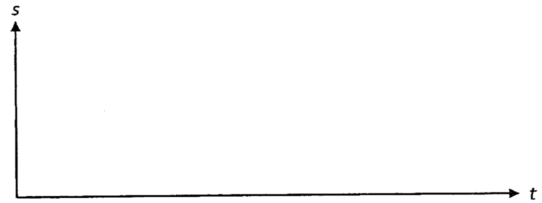
a. Describe how the acceleration of the falling object changes over the first 16 s. [2]

b. Explain how the forces acting on the falling object cause the changes in acceleration described in **a**. [2]

c. Use Fig. 4.1 to determine the acceleration of the object 5.0 s after it was released. [2]

d. Estimate the distance fallen in the first 16 s. [2]

e. Sketch the graph of distance s against time t using the axes provided below.



[2]

[Total: 10]

5. **[YJC/H2/Prelims 2018/P3/1]**

A vehicle travels from one traffic junction to another. Its acceleration-time (a-t) graph is shown in Fig. 5.1.

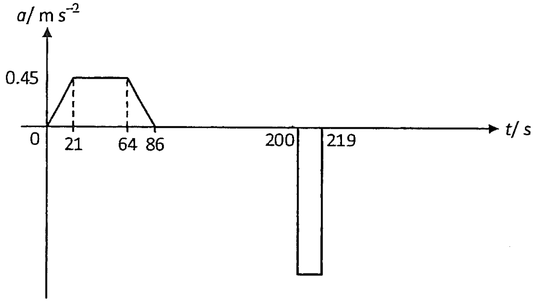


Fig. 5.1

a. Use Fig. 6.1 to determine the average acceleration of the vehicle over the first 200 s. **[2]**

b. Given that the velocity of the vehicle is 0 m s^{-1} at t = 0 s and t = 219 s, determine

i. the magnitude of its maximum velocity, and **[2]**

ii. sketch its corresponding velocity-time (v-t) graph from t = 0 s to t = 219 s in Fig 5.2.

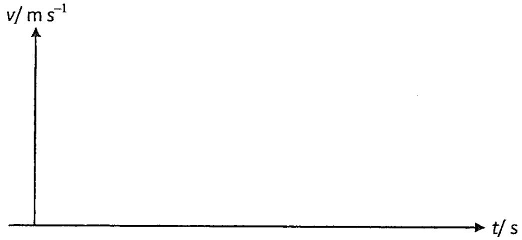


Fig. 5.2 **[4]**

[Total: 8]

6. **[DHS/H2/Prelims 2018/P3/3]**

a. i. Define *force*. **[1]**

ii. State Newton's third law of motion. **[2]**

b. Fig. 6.1 shows the variation with time t of a jumping flea's acceleration a. The acceleration a is measured in unit of g, the acceleration of free fall. The flea of mass 210 μg jumped at nearly vertical take-off angle from ground.

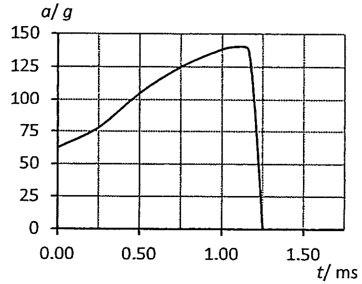


Fig. 6.1

i. Use Fig. 6.1 to

1. determine the maximum net external force acting on the jumping flea, **[2]**

2. estimate the maximum speed achieved by the flea. **[3]**

* ii. State and explain whether linear momentum is conserved during the take-off of the flea from the ground. **[2]**

[Total: 10]

7. **[EJC/H2/Prelims 2018/P2/1]**

a. State Newton's three laws of motion. **[3]**

b. Using the appropriate laws of motion, answer the question for each given situation.

i. A passenger claimed that he was sitting in the middle of a bus that was moving forward. The driver suddenly applied the brakes and a suitcase that was in the front of the bus flew backwards and hit him.
State and explain if his claim is valid. **[2]**

ii. A labourer was tasked with pulling a cart. He reasoned that whatever he exerts on the cart, the cart will exert an equal and opposite force on him. The forces will cancel out and hence it is pointless for him to pull the cart as both he and the cart will not be able to move.
State and explain if his reasoning is correct. **[2]**

iii. A man is stranded in the middle of a frozen lake with a heavy bag of gold. As there is no friction between him and the surface of the ice, he is unable to crawl back to shore.
State and explain the action that he can take to get back to shore. **[2]**

*c. A small sedan car makes a head-on collision with a large truck.

i. State and explain if the force experienced by the car is different from that experienced by the truck. **[2]**

ii. Both drivers are securely fastened to their vehicle seats. State and explain if the driver of the truck is likely to experience more severe injuries as compared to the driver of the car. **[3]**

[Total: 14]

8. [IJC/H2/Prelims 2018/P2/1 Modified]

A spring is attached at one end to a fixed point and hangs vertically with a cube attached to the other end. The cube is initially held so that the spring has zero extension, as shown in Fig. 8.1.

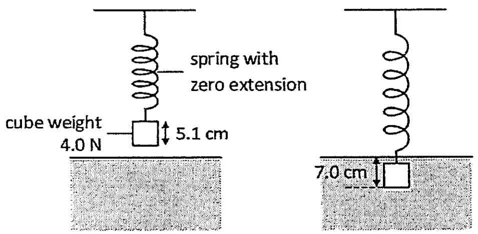


Fig. 8.1 **Fig. 8.2**

The cube has weight 4.0 N and sides of length 5.1 cm. The cube is released and sinks into water as the spring extends. The cube reaches equilibrium with its base at a depth of 7.0 cm below the water surface, as shown in Fig. 8.2.

a. The upthrust acting on the cube is 1.3 N.

i. Calculate the force exerted on the spring by the cube when it is in equilibrium in the water. [2]

ii. The spring obeys Hooke's law and has a spring constant of 30 N m^{-1}.
Determine the initial height above the water surface of the base of the cube before it was released. [2]

b. The cube in the water is released from the spring.

i. State Newton's second law of motion. [2]

ii. Determine the initial acceleration of the cube. [2]

iii. Describe and explain the variation, if any, of the acceleration of the cube as it sinks further into the water. [2]

[Total: 14]